

Problem D. Indian Summer

Time Limit 2000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

Indian summer is such a beautiful time of the year! A girl named Alyona is walking in the forest and picking a bouquet from fallen leaves. Alyona is very choosy — she doesn't take a leaf if it matches the color and the species of the tree of one of the leaves she already has. Find out how many leaves Alyona has picked.

Input

The first line contains an integer n ($1 \leq n \leq 100$) — the number of leaves Alyona has found. The next n lines contain the leaves' descriptions. Each leaf is characterized by the species of the tree it has fallen from and by the color. The species of the trees and colors are given in names, consisting of no more than 10 lowercase Latin letters. A name can not be an empty string. The species of a tree and the color are given in each line separated by a space.

Output

Output the single number — the number of Alyona's leaves.

Examples

Input	Output
5 birch yellow maple red birch yellow maple yellow maple green	4

Input	Output
3 oak yellow oak yellow oak yellow	1